

The Use of Spline Interpolation in Semi-Lagrangian Transport Models

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ABSTRACT

The accuracy of interpolating semi-Lagrangian (SL) discretization methods depends on the choice of the interpolating function. Results from barotropic transport simulations on the sphere are presented, using either bicubic Lagrangian or bicubic spline SL discretization. The spline-based scheme is shown to generate excessively noisy fields in these simulations. The two methods are then tested in a one-dimensional advection problem. The damping and dispersion relations for the schemes are examined. The analysis and numerical experiments suggest that the excessive noise found in the spline-based simulations is a consequence of insufficient damping of the small scales for small and near-integer values of the Courant number. Inspection of the local Courant number for the two-dimensional spline-based simulation confirms this hypothesis. This noise can be controlled by adding a scale-selective diffusion term to the spline-based scheme, while retaining its excellent dispersion characteristics.

1. Introduction

In recent years semi-Lagrangian (SL) methods have come to be widely used in numerical weather prediction and climate modeling. One reason for this is computational efficiency, since these methods generally allow a longer time step than Eulerian discretizations. Another reason is the natural way in which the advection equation is expressed in this framework (e.g., Bates et al. 1995).

Modern data assimilation methods rely on the tangent linear and adjoint versions of a given model for propagating information in time, and it is thus important to ensure that the discrete model is actually differentiable with respect to the model state variables [e.g., Polavarapu et al. (1996), henceforth P96]. For interpolating SL schemes, differentiability with respect to the state is equivalent to differentiability across grid points of the interpolant used (P96). Spline interpolants are constructed to be differentiable across grid points, while the Lagrangian polynomial interpolants often used in SL schemes are not differentiable. From the point of

view of data assimilation, spline interpolation therefore appears to be advantageous.

It has long been known that SL schemes using cubic spline interpolation on a Cartesian grid are both less diffusive and less dispersive than corresponding algorithms using cubic Lagrangian interpolation (Purnell 1976; Pudykiewicz and Staniforth 1984). Makar and Karpik (1996, henceforth MK96) compared different interpolation methods for SL transport modeling on the sphere and found that a cubic B-spline algorithm yielded superior results in terms of mass conservation, shape similarity, location, and preservation of maxima and minima by most of the scoring methods that they used. Huang (1994), on the other hand, showed that a cubic spline algorithm may lead to severe over- and undershooting in a deformation test.

Here we present results from a series of tests in which SL schemes based on cubic spline and cubic Lagrangian interpolation are compared in one and two dimensions. Results from the two-dimensional tests show that for realistic applications, cubic spline algorithms may lead to the generation of spurious small-scale features. The one-dimensional tests show that the cubic spline algorithm may generate substantial over- and undershooting even for flows that are much simpler than the deformation field used by Huang (1994) and that its behavior is very sensitive to the Courant number of the flow.

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Such over- and undershooting can be remedied by imposing a monotonicity condition [e.g., Bermejo and Staniforth (1992)]. However, this generally leads to a loss of differentiability and is therefore not considered here.

We revisit the dispersion relations for SL schemes using either cubic spline or cubic Lagrangian interpolation, for the one-dimensional case with a uniform advecting wind field. We show that the spurious behavior of the spline algorithm is caused by a combination of little damping and nonnegligible dispersion at the smallest scales for small values of the nondimensional velocity (Courant number). Finally we examine the possibility of improving the spline algorithm by adding a diffusion term, similar to that proposed for a Cartesian grid by Pudykiewicz and Staniforth (1984).

2. A two-dimensional model example

In this section we present a comparison between bicubic spline interpolation and bicubic Lagrangian interpolation for a semi-Lagrangian barotropic model on the sphere. The model differential equations are

$$\frac{d(\zeta + f)}{dt} = 0, \quad (1)$$

$$\frac{d\mu}{dt} = 0, \quad (2)$$

where d/dt is the material derivative, f is the Coriolis parameter, $\zeta = \nabla^2\Psi$ is the relative vorticity, Ψ is the streamfunction, and μ is a passively advected quantity. The model thus consists of a nonlinear equation (1) governing the flow, coupled with a linear advection equation (2). This system has been studied previously by Riishøjgaard (1996) for the problem of wind retrieval from constituent observations. In the examples shown here, $\eta = \zeta + f$ and μ are initialized so as to resemble absolute vorticity and ozone mixing ratio, respectively, at about 200 hPa.

The standard SL time discretization of (1) is

$$(\zeta + f)_{k,*}^{n+1} = (\zeta + f)_{k,*}^n. \quad (3)$$

Here the subscript $(k,*)$ indicates the departure point at time t^n corresponding to the grid point k at time t^{n+1} . To avoid the Rossby wave instability associated with this discretization we follow the approach of Bates et al. (1995) in expanding the Coriolis parameter of the departure point in a Taylor series around its value f_0 at the arrival point. This leads to the following elliptic equation for the streamfunction at time t^{n+1} :

$$\nabla^2\Psi^{n+1} + \frac{\beta_0\Delta t}{2a\cos\theta} \frac{\partial\Psi^{n+1}}{\partial\lambda} = r^n, \quad (4)$$

where $\beta_0 = \partial f_0/\partial\theta$, a is the radius of the earth, and θ is latitude. The expression for r^n is

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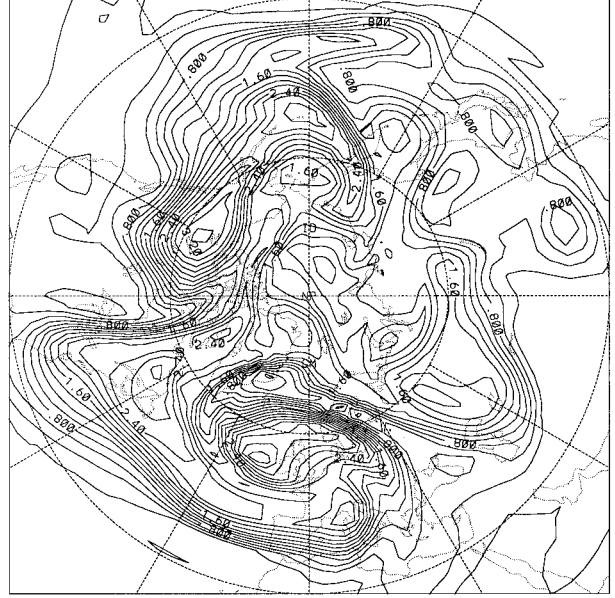


FIG. 1. Initial constituent field for the barotropic model simulations, shown in a polar stereographic plot of the Northern Hemisphere north of 30°N.

$$r^n = \zeta_*^n - \frac{\beta_0\Delta t}{2}v_*^n - \frac{f_0}{2}\left(\frac{v_*^n\Delta t}{a}\right)^2. \quad (5)$$

The forcing function r^n is thus a sum of three terms, in which ζ^n and v^n are evaluated at the departure points. The procedure for locating the departure points is similar to that of Bates et al. (1995). The elliptic equation is solved using the algorithm of Moorthi and Higgins (1992).

The time discretization of (2) is straightforward:

$$(\mu)_k^{n+1} = (\mu)_{k,*}^n. \quad (6)$$

Each departure point for μ is the same as that for ζ^n and v^n , so the solution of (2) just amounts to the interpolation of an extra field to the set of departure points. Our emphasis here will be on the interpolation method used for this evaluation once the departure points have been found, and we will concentrate on the evolution of the passive variable μ .

The model equations are solved numerically over a uniform grid on the sphere with 65 latitudes and 128 longitudes; the time step is $\Delta t = 1800$ s. The initial fields were taken from a reference simulation with a three-dimensional climate model described in Riishøjgaard (1996). In Fig. 1 we show the initial distribution of μ over the Northern Hemisphere poleward of 30°N. Even though it is model generated, the field has most of the main features of a real ozone field. An important aspect from the point of view of the transport algorithm is that the field contains large areas of nearly constant values separated by narrow zones of sharp gradients.

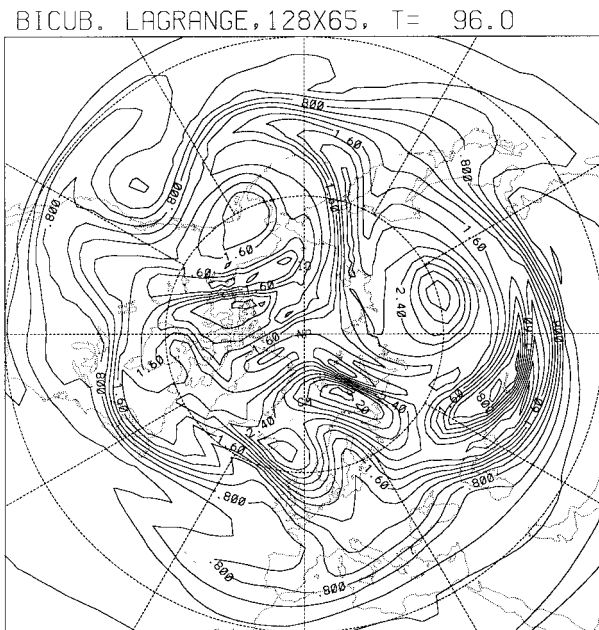


FIG. 2. Constituent field after 96 h of simulation with all fields evolved using bicubic Lagrangian interpolation.

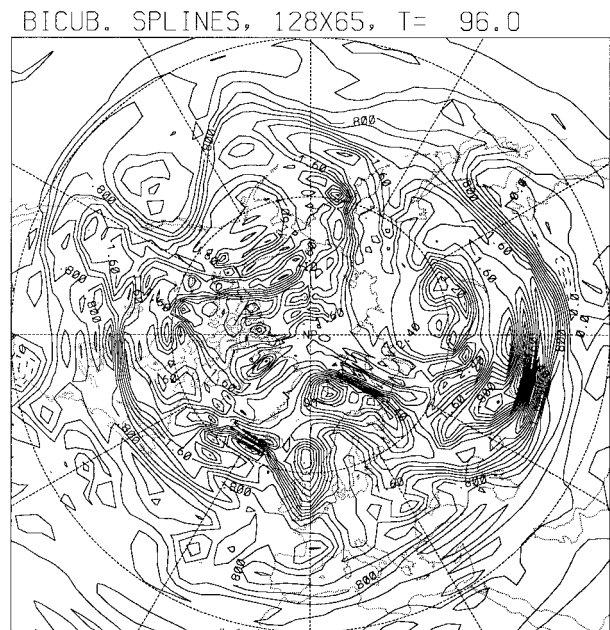


FIG. 3. Constituent field after 96 h of simulation with all fields evolved using bicubic B-spline interpolation.

In Fig. 2 the field is shown after 96 h of simulation, with r^n and μ^n evaluated at the departure point locations using bicubic Lagrangian interpolation. The scheme seems to behave reasonably well. No new features are generated, and no major over- and undershooting is evident. The scheme does seem somewhat dissipative: the gradients are less sharp in Fig. 2 than in Fig. 1.

Figure 3 shows the same field as in Fig. 2, but with both r^n and μ^n evaluated at each time step using the bicubic B-spline algorithm of MK96. While the same features can be distinguished in the two simulations, the B-spline scheme appears much noisier than the bicubic Lagrangian scheme. After 96 h numerous new local extrema have been generated, and the global maximum and global minimum are both outside the range of the initial field. The latter phenomenon violates the advection equation (2). On the other hand, the scheme does seem better at maintaining the gradients of the field.

Both because of the aforementioned differentiability of spline interpolation (P96), and because the spline scheme has been shown to perform well in simpler tests (MK96), it is of interest to examine why the B-spline scheme performs so poorly here. We have compared the results shown in Fig. 3 with those of an independently coded classic bicubic spline algorithm, and the fields were essentially identical. We have also repeated the cone advection experiments of MK96, again obtaining results similar to theirs.

We have furthermore tested the impact of the spline algorithm on equations (1) and (2) separately, by way of two experiments:

- B-spline interpolation of r ; bicubic Lagrangian interpolation of μ ;
- bicubic Lagrangian interpolation of r ; B-spline interpolation of μ .

Results from these experiments (not shown) indicate that the quality of the tracer field μ is determined by the interpolation method used for evaluating the field itself, while it is largely unaffected by the interpolation chosen for the forcing function r . In the following section we therefore investigate further the SL discretization of advection equation (2) using the cubic spline and cubic Lagrangian interpolation schemes.

3. Dispersion and damping in two semi-Lagrangian schemes

In this section we present first a comparative Fourier analysis of two SL discretizations of (2) with constant advection speed u on a one-dimensional 2π -periodic domain, along with sample results from numerical experiments in one dimension. Separate analyses of the two schemes have been presented by Pudykiewicz and Staniforth (1984) and McDonald (1984) for the cubic Lagrangian interpolation, and by Purnell (1976) for the cubic spline interpolation. The discretized equation is

$$\mu_k^{n+1} = \tilde{\mu}^n(x_k - u\Delta t), \quad (7)$$

where $\tilde{\mu}^n$ is the interpolant through the field at time t^n , and $x_k - u\Delta t$ is the departure point corresponding to the arrival point x_k at time t^{n+1} . The grid is uniformly spaced, $x_k = x_0 + k\Delta x$, and we will assume that the Courant number $C = |u|\Delta t/\Delta x$ is between 0 and 1, and

that $u \geq 0$, so that the departure point $x_k - u\Delta t = x_k - C\Delta x$ corresponding to the arrival point x_k is found in the interval (x_{k-1}, x_k) .

We present results for cubic Lagrangian interpolation and for cubic spline interpolation. Both of these can be written in the form (as in P96)

$$\begin{aligned} \tilde{\mu}_k^n(x_k - C\Delta x) \\ = C\mu_{k-1}^n + (1 - C)\mu_k^n + \frac{1}{6}(C^3 - C)(\Delta x)^2\mu_{k-1}^{n''} \\ + \frac{1}{6}[(1 - C)^3 - (1 - C)](\Delta x)^2\mu_k^{n''}, \end{aligned} \quad (8)$$

where $\mu_k^{n''}$ is a discrete approximation to the second derivative of the field with respect to the horizontal coordinate. The difference between the two interpolation schemes lies in the way these second derivatives are approximated.

The cubic Lagrangian interpolant is obtained by letting

$$\mu_k^{n''} = \frac{\mu_{k-1} - 2\mu_k + \mu_{k+1}}{(\Delta x)^2}. \quad (9)$$

By inserting (9) into (8) one can verify that this interpolant can be written as a weighted sum of the values at the four nearest grid points:

$$\tilde{\mu}_k^n(x_k - C\Delta x) = g_{-2}\mu_{k-2}^n + g_{-1}\mu_{k-1}^n + g_0\mu_k^n + g_1\mu_{k+1}^n, \quad (10)$$

with the weights given by

$$\begin{aligned} g_{-2} &= \frac{C^3 - C}{6} \\ g_{-1} &= \frac{-C^3 + C^2 + 2C}{2} \\ g_0 &= \frac{C^3 - 2C^2 - C + 2}{2} \\ g_1 &= \frac{-C^3 + 3C^2 - 2C}{6}. \end{aligned} \quad (11)$$

The resulting SL scheme is obtained by substituting (10) into (7).

The cubic spline interpolant is derived by requiring that its first derivative be differentiable at the grid points. This leads to the following system of equations for the (continuous) second derivatives [e.g., Press et al. (1992)]:

$$\frac{\mu_{k-1}^{n''}}{6} + \frac{2\mu_k^{n''}}{3} + \frac{\mu_{k+1}^{n''}}{6} = \frac{\mu_{k-1} - 2\mu_k + \mu_{k+1}}{(\Delta x)^2}. \quad (12)$$

By evaluating the sum $(1/6)\mu_{k-1}^{n+1} + (2/3)\mu_k^{n+1} + (1/6)\mu_{k+1}^{n+1}$, using (7), (8), and (12), one can verify that for the cubic spline interpolant, the SL time stepping is given by the following system of equations:

$$\begin{aligned} \frac{1}{6}\mu_{k-1}^{n+1} + \frac{2}{3}\mu_k^{n+1} + \frac{1}{6}\mu_{k+1}^{n+1} \\ = s_{-2}\mu_{k-2}^n + s_{-1}\mu_{k-1}^n + s_0\mu_k^n + s_1\mu_{k+1}^n, \end{aligned} \quad (13)$$

where the coefficients are

$$\begin{aligned} s_{-2} &= \frac{C^3}{6} \\ s_{-1} &= C + \frac{(1 - C)^3 - 2C^3}{6} \\ s_0 &= (1 - C) + \frac{C^3 - 2(1 - C)^3}{6} \\ s_1 &= \frac{(1 - C)^3}{6}. \end{aligned} \quad (14)$$

Both the cubic Lagrangian scheme (10) and the cubic spline scheme (13) can thus be written in matrix form

$$\mathbf{M}\mu^{n+1} = \mathbf{S}\mu^n, \quad (15)$$

where $\mathbf{M}$ and $\mathbf{S}$ are circulant matrices [e.g., Davis (1979)]. The discrete Fourier transform (DFT) of (15) is

$$\hat{\mu}^{n+1} = \hat{\Psi}\hat{\mu}^n, \quad (16)$$

where $\hat{\mu}$ is the DFT of μ , $\hat{\Psi} = \hat{\mathbf{S}}\hat{\mathbf{M}}^{-1}$, and the diagonal matrices $\hat{\mathbf{S}}$ and $\hat{\mathbf{M}}$ are the DFTs of $\mathbf{S}$ and $\mathbf{M}$, respectively.

For the cubic Lagrangian scheme, $\mathbf{M}$ is the identity matrix, so that $\hat{\Psi} = \hat{\mathbf{S}}$. It follows from (10) that in this case the diagonal elements of the diagonal matrix $\hat{\Psi}$, indexed by the spatial wavenumber ξ , are given by (e.g., Sod 1985)

$$\hat{\Psi}(\xi\Delta x) = g_{-2}e^{-2i\xi\Delta x} + g_{-1}e^{-i\xi\Delta x} + g_0 + g_1e^{i\xi\Delta x}, \quad (17)$$

$\hat{\Psi}(\xi\Delta x)$ is also known as the symbol of the scheme (15). For the cubic spline scheme (13) we have

$$\hat{\Psi}(\xi\Delta x) = \frac{s_{-2}e^{-2i\xi\Delta x} + s_{-1}e^{-i\xi\Delta x} + s_0 + s_1e^{i\xi\Delta x}}{2/3 + 1/3 \cos(\xi\Delta x)}. \quad (18)$$

We can now examine the damping and dispersion characteristics of the schemes using (17) and (18). The amplification factor of a given scheme is $|\hat{\Psi}(\xi\Delta x)|$, while the dispersion relation is

$$\nu = \frac{-1}{u\Delta t} \arg[\hat{\Psi}(\xi\Delta x)]. \quad (19)$$

In comparison, the original differential equation $\mu_t + u\mu_x = 0$ has Fourier transform $\hat{\mu}_t + i\xi u\hat{\mu} = 0$. The solution of this equation over one time step is

$$\hat{\mu}^{n+1} = e^{-i\xi u\Delta t} \hat{\mu}^n. \quad (20)$$

That is, it has unit amplification factor and its dispersion relation is $\nu = \xi$.

The amplification factor and the dispersion $\nu\Delta x$ are plotted in Figs. 4a and 4b, respectively, as functions of the nondimensional spatial frequency $\xi\Delta x$, for the

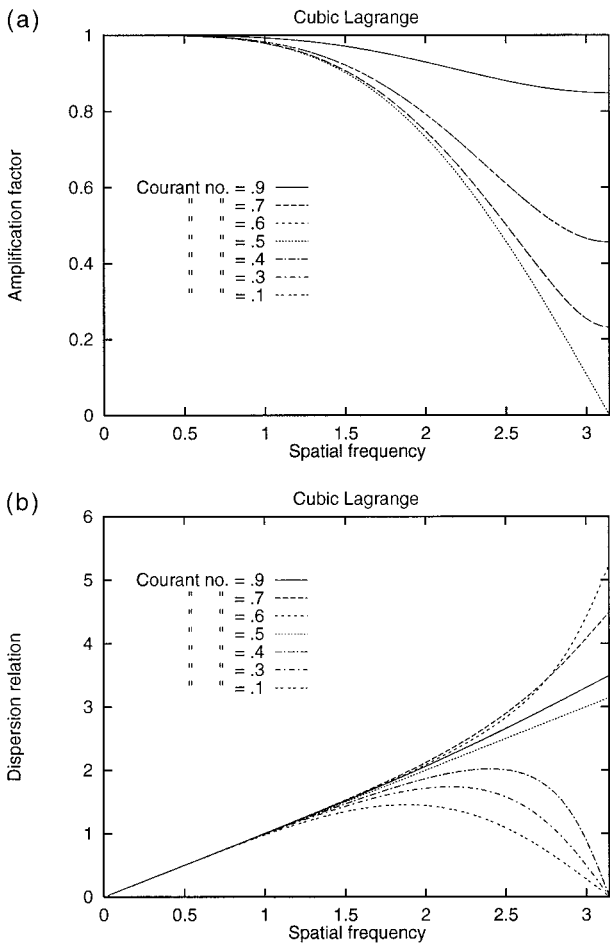


FIG. 4. Amplification factor (a) and dispersion relation (b) for semi-Lagrangian discretization of the 1D advection equation with cubic Lagrangian interpolation. The dispersion plot shows $v\Delta x$ vs $\xi\Delta x$.

scheme based on cubic Lagrangian interpolation. Curves for several values of the Courant number C are shown. Figure 4a shows that the scheme is most diffusive for $C = 0.5$, where the $2\Delta x$ wave is completely eliminated within one time step. The algorithm is moderately diffusive for C close to either 0 or 1. Note that the amplification factor for a given C is equal to that for $1 - C$, and that some of the curves in Fig. 4a therefore coincide. Figure 4b shows that for a Courant number of 0.5 there is no spurious dispersion. For all Courant numbers greater than 0.5 the numerical solution leads the analytical solution, and for values less than 0.5 the numerical solution lags the analytical solution. As C approaches 0 an increasing part of the spectrum exhibits substantial phase lags.

Plots analogous to those of Fig. 4 are shown in Fig. 5 for the cubic spline scheme. Also for this scheme, the $2\Delta x$ wave is completely eliminated within one time step for $C = 0.5$. However, the curves in Fig. 5a stay close to a value of 1 over much of the spectrum, before falling off much more steeply toward the smallest scales than

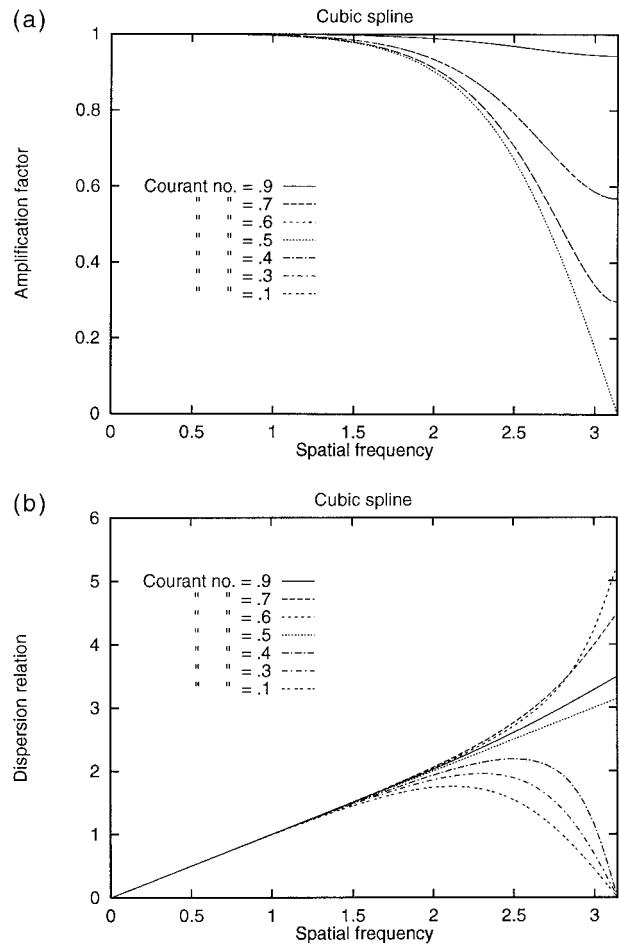


FIG. 5. As in Fig. 4 but for cubic spline interpolation.

in Fig. 4a. The spline-based scheme is thus less damping than the cubic Lagrangian scheme. In Fig. 5b we see that the dispersive character of the cubic spline scheme is similar to, but less severe than, that of the cubic Lagrangian scheme shown in Fig. 4b.

That schemes based on cubic splines are both less dispersive and less dissipative than cubic Lagrangian-based schemes has generally been regarded as an advantage of spline-based schemes. However, the combination of little damping and nonnegligible dispersion at the smallest scales can actually be problematic, even for very simple flows. In order to illustrate this, we show examples of numerical advection experiments on a one-dimensional 2π -periodic domain with constant C , using the two different SL discretizations analyzed above.

In the first example the initial structure consists of a wave with period $4\Delta x$; the Courant number is $C = 0.25$. In Fig. 6 the numerical solution for each of the schemes is shown together with the analytical solution (a simple translation of the initial structure) after 40 time steps over a small part of the domain. This figure confirms that the cubic spline scheme is both less damping and less dispersive than the cubic Lagrangian scheme. After

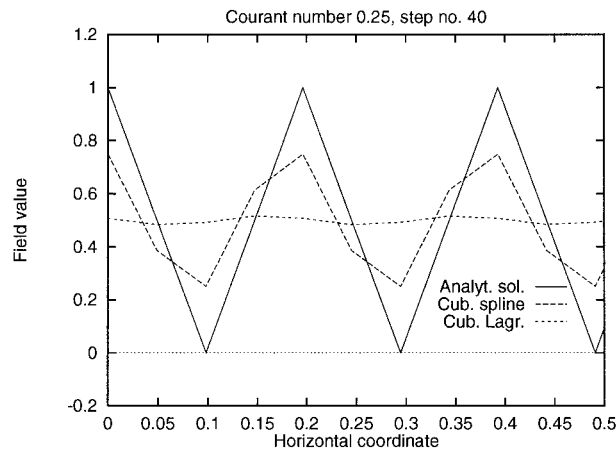


FIG. 6. Advection of a $4\Delta x$ wave with Courant number 0.25. The analytical solution is shown together with two different numerical solutions after 40 time steps. Only a small part of the domain is shown.

40 time steps the cubic Lagrangian solution is roughly 90° out of phase with the analytical solution, and the amplitude is reduced to about 3% of the original value. The phase of the cubic spline solution only lags that of the analytical solution by about 30%, but it still has more than 50% of the initial amplitude left.

The next example is advection in one dimension of a structure composed of many different wavenumbers. The initial distribution is boxlike, taking on a value of 1 over the interval $(1, 1 + \pi)$ and 0 everywhere else. This is a severe test of the behavior of the schemes in zones with steep gradients, similar to what is often seen in water vapor or ozone mixing ratio fields for instance. We show plots for two values of the Courant number 40 time steps into the simulation. One may think of these as representing two different flow speeds in a realistic field, where the velocity varies in time and space.

In Fig. 7 the two numerical solutions are shown together with the analytical solution for $C = 0.25$. Only half of the domain is shown. The cubic Lagrange algorithm has smoothed out the step more than the spline algorithm, but it relaxes back to the correct value on both sides of the step with only moderate over- and undershooting. The result of the spline algorithm is noisier, with several oscillations in the vicinity of the transition. As expected from the dispersion relations, the shortwave components tend to lag those of the analytical solution.

In Fig. 8 corresponding results are shown for $C = 0.05$. As suggested by Figs. 4a and 5a, the small scales are less damped here than in the $C = 0.25$ case. That is, the steep transition is better preserved. It is also evident, however, that dispersion of these higher frequencies make them appear as unwanted noise in regions where the analytical solution is constant, especially for the spline-based scheme, since it is less diffusive than the cubic Lagrangian scheme.

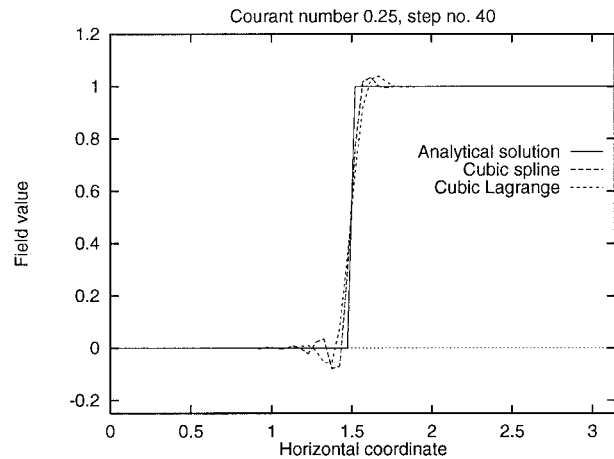


FIG. 7. Advection of a rectangular wave with Courant number 0.25. The analytical solution is shown together with two different numerical solutions after 40 time steps.

These examples demonstrate the potential for serious problems with spline-based schemes at small Courant numbers when used to advect fields with steep gradients.

We note that the velocities u and $-u$ both lead to the same Courant number C . So far we have assumed that u is positive. By symmetry, the numerical solution for a negative $u = -C\Delta x/\Delta t$ will be similar to the one for a positive $u = C\Delta x/\Delta t$ (e.g., Fig. 8), with the small-scale noise on the opposite side of the step. The solution for Courant number $(1 - C)$ is identical to the solution for Courant number $-C$, only shifted n grid cells to the right, where n is the number of time steps. In general, since both schemes are exact for integer Courant numbers, the noise pattern for a given value of C will be found for all $m \pm C$, where m is a positive integer.

While the maximum Courant number for realistic two- or three-dimensional atmospheric flows can largely be controlled by the experimenter, the minimum Courant number generally cannot, since near-zero velocities are bound to occur from time to time. Semi-Lagrangian

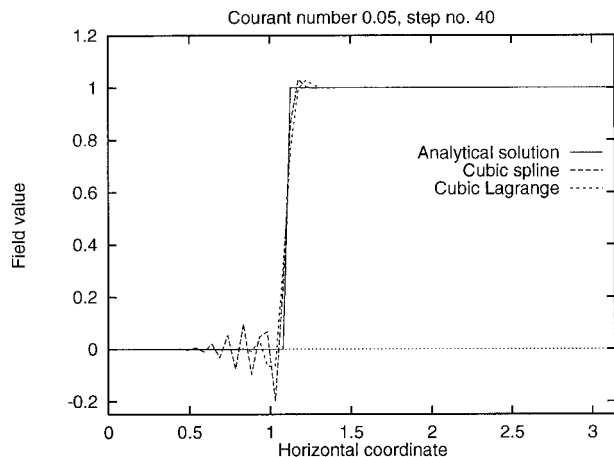


FIG. 8. As in Fig. 7 but with a Courant number of 0.05.

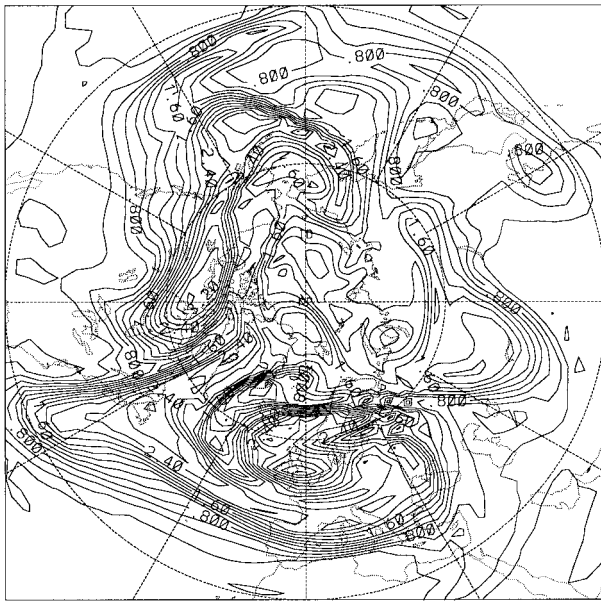


FIG. 9. Constituent field after 12 h of simulation with all fields evolved using bicubic B-spline interpolation.

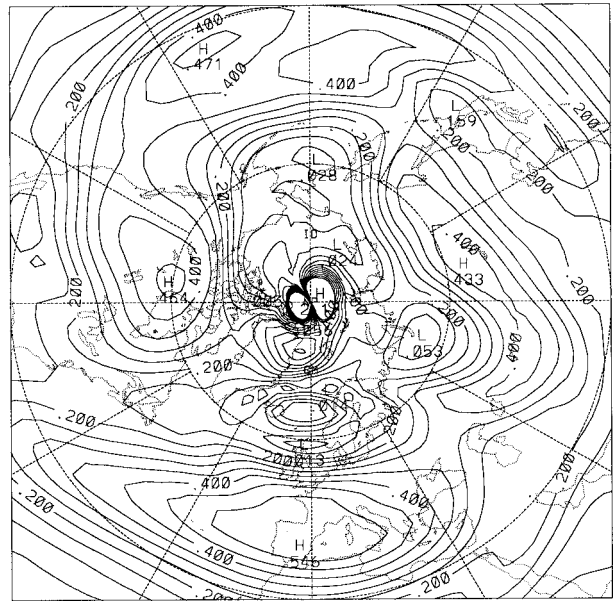


FIG. 10. Local Courant number of flow after 12 h of simulation (cf. Fig. 9).

schemes have become popular in part because of their stability, a property that permits the use of large Courant numbers (large time steps). The potential problems associated with small Courant numbers appear to have received little attention.

We now show that the excessive noise in the two-dimensional cubic spline simulation discussed in the previous section (Fig. 3) is likely to have been caused by the inability of the spline-based scheme to handle correctly the combination of small Courant numbers and sharp gradients of the field. Figure 9 shows the tracer field from the same simulation as in Fig. 3, but at an earlier stage in the run (12 h). At this time the noise is beginning to become noticeable. A train of presumably unphysical oscillations originating over northern Scandinavia extends toward Russia, and a similar train stretches from eastern Siberia down across the Pacific. Both of these features are associated with large gradients in the field. Another area with a sharp gradient over eastern Greenland shows signs of developing small-scale features. On the other hand, there are other areas with large gradients, for instance, over Canada and over the Mediterranean, that are not affected by noise at this time.

For comparison we show the local Courant number of the flow, defined here as $C_i \equiv \max(|u|\Delta t/\Delta x, |v|\Delta t/\Delta y)$, in Fig. 10. It is seen that C_i has very low values over northern Scandinavia, with a minimum of 0.014 off the coast of Norway. The area of low values curves down along eastern Greenland. Another patch of low C_i , with a minimum value of 0.028, is found over eastern Siberia. These regions of low C_i correspond closely to the disturbances discussed above. On the other hand, values approaching 0.5 prevail over the Mediterranean

and over Canada, where the large gradients do not lead to spurious oscillations in Fig. 9.

The relationship between Figs. 9 and 10, which is typical for those of other validation times within the first 36 h of the model trajectory (not shown), is evidence that the combination of small Courant numbers and large gradients is indeed the cause of the noise in the spline-based solution. At ranges beyond 36 h the advection tends to have separated the region of error from the region in which it originated, and a direct matching of features is no longer possible.

4. A spline-based scheme with diffusion

In the previous section it was shown that the problems seen when spline interpolation is used for SL transport are likely to be caused by the fact that for small Courant numbers the scheme has little dissipation but nonnegligible dispersion in the smallest spatial scales. Thus, even though the spline scheme is the less dispersive of the schemes examined here, its relatively nondissipative character can actually bring the dispersive scales into focus in an unfortunate way. Therefore, it is natural to attempt to cure this problem by adding some form of diffusion to the scheme. This will leave the dispersion unchanged, while a select range of spatial scales can be damped with a time constant of choice. Although there may be theoretical and empirical justification for applying some kind of spatial filtering to the scheme, both the choice of the filter itself and the time constant with which it is applied remain somewhat arbitrary.

We present results obtained using fourth-order semi-implicit diffusion, first for the 1D problem and then for the 2D coupled model. In the first case, the experimental

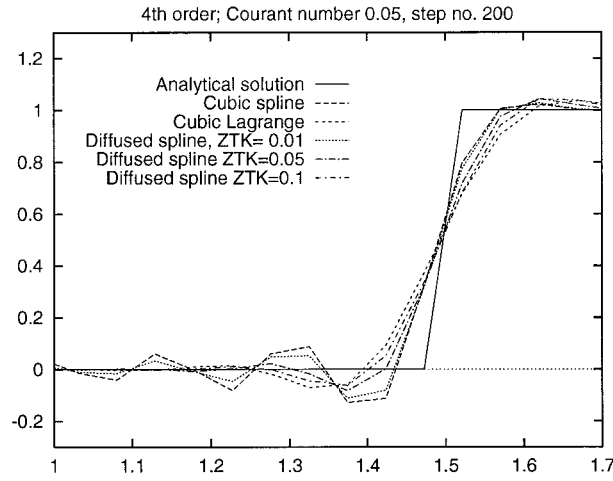


FIG. 11. Advection of a rectangular wave with Courant number 0.05. The analytical solution is shown together with results of the cubic Lagrangian scheme, the cubic spline scheme, and the cubic spline scheme with fourth-order diffusion. Results are shown after 200 time steps.

framework is the constant Courant number linear periodic advection of a square wave discussed in the previous section (Figs. 7 and 8). Preliminary experiments (not shown) indicated that second-order diffusion damped out the small scales excessively.

We introduce diffusion in the spline algorithm by time-splitting, solving first the discrete advection equation (7) using the spline for evaluating the right-hand side. Let this intermediate solution be denoted by $\hat{\mu}$. We then seek the solution μ to the fourth-order diffusion equation

$$\frac{\partial \mu}{\partial t} = -K \nabla^4 \mu, \quad K > 0, \quad (21)$$

using the semi-implicit discretization

$$\frac{\mu^{n+1} - \hat{\mu}}{\Delta t} = -K \frac{1}{2} \left(\frac{\partial^4 \mu^{n+1}}{\partial x^4} + \frac{\partial^4 \hat{\mu}}{\partial x^4} \right). \quad (22)$$

This equation is solved in the spectral domain, using the fact that the complex exponentials are eigenfunctions of the operator ∇^4 . Since the symbol of the spectral scheme for (22) is

$$\hat{\Psi}(\xi \Delta x) = \frac{1 - \xi^4 K(\Delta t/2)}{1 + \xi^4 K(\Delta t/2)}, \quad (23)$$

and the maximum spatial frequency on the discrete grid is $\xi = \pi/\Delta x$, the overall time-split scheme is stable and damping provided that

$$\frac{K \Delta t}{2} \left(\frac{\pi}{\Delta x} \right)^4 \leq 1. \quad (24)$$

In Fig. 11 results are shown for three different values of the diffusion coefficient K . For reference we also show the analytical solution along with results of the

original cubic spline and cubic Lagrangian algorithms. Only a small portion of the domain is shown in order to focus on the behavior in the vicinity of the step in the distribution. The results are plotted after 200 time steps. The diffusion parameter shown in the plot is the relative amount $1 - \alpha$ by which the amplitude of shortest wave is reduced per time step, where

$$\alpha = \frac{1 - K(\Delta t/2)(\pi/\Delta x)^4}{1 + K(\Delta t/2)(\pi/\Delta x)^4}. \quad (25)$$

It is seen that fourth-order diffusion helps control the dispersive region without excessively smoothing the gradient in the zone of interest. In this example a diffusion parameter $1 - \alpha \approx 0.1$ seems adequate. We concede that no firm conclusion should be drawn from this single example. However, numerous experiments with other initial conditions and other Courant numbers (not shown) support the hypothesis that to some extent the problems with the spline algorithm can indeed be cured by applying fourth-order diffusion. The positive aspects of the spline scheme in terms of overall accuracy seem to prevail.

We now return to the two-dimensional problem. In view of the results obtained by adding a semi-implicit diffusion term to the spline algorithm in one dimension, it was decided to test this approach also in the two-dimensional coupled vorticity-transport model. We therefore implemented the fourth-order horizontal diffusion algorithm for gridpoint models on the sphere described by Li et al. (1994). A range of different diffusion coefficients were tested, and we present only a single example here. We have tested the diffusion algorithm both in the tracer advection equation alone and in the full model system. Since results with the latter were found to be marginally better, we only show results from this case here.

In Fig. 12 the tracer field is shown after 96 h of simulation with fourth-order diffusion. In this example the diffusion coefficient was chosen so that the amplitude of a $2\Delta x$ wave at 60°N would be reduced by a factor of 0.1 per time step. By trial and error it was found that this was the smallest diffusion coefficient that would keep the values of the field within their initial range throughout the course of the simulation. A comparison of the spline-diffusion algorithm with the cubic Lagrangian algorithm (Fig. 2) shows that the results are in fact similar, with a slight advantage for the spline-diffusion scheme. The latter seems better at creating elongated features drawn out by the flow, has a sharper gradient in certain areas, and is actually now less noisy than the cubic Lagrangian scheme.

With the caveat that we do not have an analytical solution at hand for reference in this type of experiment, this example nevertheless seems to confirm what was shown for one dimension, namely, that a spline-diffusion scheme retains some of the best characteristics of the cubic spline and cubic Lagrangian interpolation

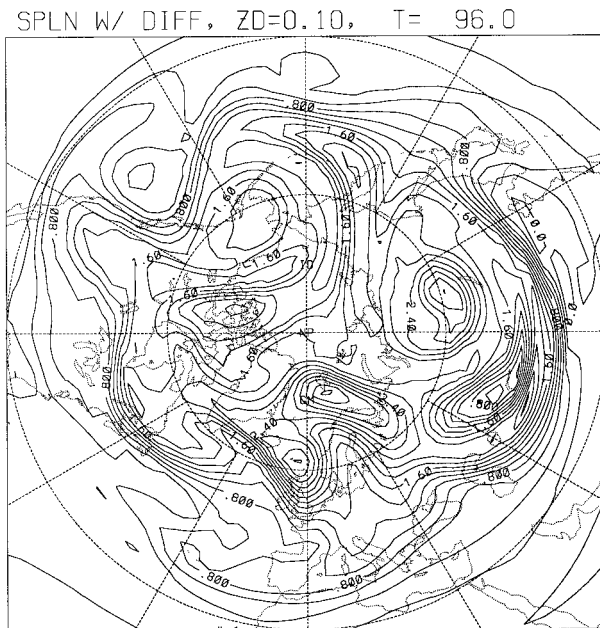


FIG. 12. As in Fig. 3 but with fourth-order diffusion added to the scheme.

schemes. In terms of computational cost the original cubic spline scheme is 34% more expensive than the cubic Lagrangian scheme, and the corresponding figure for the spline-diffusion scheme is 75%. These figures are from a research implementation of the various algorithms in the two-dimensional model. No attempt was made to optimize their computational efficiency.

5. Summary and conclusions

We have shown that an interpolating semi-Lagrangian scheme using bicubic splines leads to excessively noisy constituent fields in a barotropic model aimed at simulating atmospheric dynamics and tracer transport at about the tropopause level. In order to find the cause of this problem we have examined semi-Lagrangian algorithms based on cubic Lagrangian and cubic spline interpolation in one dimension, both in terms of their damping and dispersion relations, and in terms of their performance in simplified one-dimensional transport problems.

The cubic spline scheme was found to have excellent theoretical properties. However, for transport of distributions with steep gradients (i.e., substantial energy in the smaller scales) the results were contaminated by excessive noise. We have shown that this is due to the dispersive part of the spectrum (i.e., the small scales) being insufficiently damped by the scheme. The problem occurs for Courant numbers near 0 as well as for near-integer values. This means that it cannot be avoided by altering the length of the time step, since the minimum Courant number is beyond control for realistic atmospheric applications.

In a one-dimensional test case the noise could be kept under control by adding a semi-implicit fourth-order diffusion term to the scheme. The resulting scheme showed improvement over both the cubic Lagrangian scheme and the original cubic spline scheme without diffusion. The addition of fourth-order diffusion was subsequently tested also in the barotropic model framework, and results from this test confirmed the improvement over the other two schemes.

We note that it may be possible to obtain further improvement by allowing the diffusion coefficient to depend on the local Courant number. This possibility has not yet been tested.

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